

ANALYSIS - I : MTH 301 : QUIZ: 2025-26

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- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only. Maximum marks: 20 Time: 6:45 pm - 8:00 pm.

1. (a) Can we find a metric space (M, d) having exactly 2025 open sets? Justify your answer. [2.5]

(b) Let $f(x) = \log\left(\frac{23 + \sin(2\pi x)^7}{1 + e^{2x}}\right)$ and $x_n = \frac{n + \sin n}{n}$ be a sequence for $n \in \mathbb{N}$. Calculate $\limsup f(x_n)$ and $\liminf f(x_n)$. [2.5]

(a).

$$(b) f(n) = \log\left(\frac{23 + \sin(2\pi n)^7}{1 + e^{2n}}\right)$$

$$f(x_n) = \log\left(\frac{23 + \sin\left(2\pi\left(1 + \frac{\sin n}{n}\right)\right)^7}{1 + e^{2\left(1 + \frac{\sin n}{n}\right)}}\right)$$

2. Let $(\mathbb{R}, d)$ be the usual metric space and $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Show that the set $U = \{x \in \mathbb{R} : f(x) > a\}$ for some $a \in \mathbb{R}$ is an open set. Check whether U will be an open set if f is monotone but not continuous. [5]

$(\mathbb{R}, d)$ is the metric space with $d(x, y) = |x - y|$.

$f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous

$$U = \{x \in \mathbb{R} : f(x) > a\}$$

if U is empty $U = \emptyset$

$\Rightarrow U$ is open set (by defn)

if U is not empty $\Rightarrow \exists n_0 \in U$ s.t. $f(n_0) > a$.

since $f(x)$ is continuous at $x = n_0$.

$\Rightarrow \forall \epsilon > 0, \exists \delta$ s.t. $\forall x \in (n_0 - \delta, n_0 + \delta)$ $|f(x) - f(n_0)| < \epsilon$.

take $\epsilon = f(n_0) - a > 0$.

$\Rightarrow \exists \delta_0$ s.t. $\forall x \in (n_0 - \delta_0, n_0 + \delta_0)$

$$|f(x) - f(n_0)| < f(n_0) - a$$

$$\Rightarrow a < f(x) < 2f(n_0) - a$$

since $f(x) > a \Rightarrow x \in U$.

for U to be open;

$\forall x \in U, \exists \epsilon > 0$ s.t.

$$B_\epsilon(x) = \{y : |y - x| < \epsilon\}$$

belongs to the set U .

$$B_\epsilon(x) \subseteq U$$

consider U to show U is

open take $\epsilon = \delta$.

and we have that for

all $|y - x_0| < \delta$

$$f(y) > a \Rightarrow y \in U \quad f(y) > a \Rightarrow y \in U$$

~~the set is open~~ w.l.o.g $f(x)$ is non decreasing

~~the set only has open~~ the fn only has jump discontinuities as it is monotone.

consider $U = \{x \in \mathbb{R} : f(x) > a\}$

take a point in this set that is discontinuous.

~~if f is monotone but not continuous~~

~~we know that~~

~~the~~ if f is monotone all discontinuous pts are jump discontinuity.

for continuous pts in x we know $\exists B_r(x)$

~~st if belongs to x .~~

~~$B_r(x) \subseteq U$.~~

w.l.o.g assume non decreasing.

for pts at discontinuity $f(c^-) < f(c^+)$

~~we can find.~~ since f is

~~monotone~~

~~not open~~ part 2: the set is not open

consider since monotone

wlog take non decreasing

~~take~~ only jump discontinuity and right continuous

$f(c^-) < f(c^+)$ for continuous

take $a > f(c^-) < f(c)$

~~$f(c^+) < a < f(c)$~~

we know that $f(c^-) < a < f(c) < f(c^+)$

take a s.t. $f(c^-) < a < f(c) < f(c^+)$

but we know $\exists$ no $n \in \mathbb{R}$ s.t. $f(n) < a < f(c) \Rightarrow$ all n close to c don't lie in U

it is enough work or not? Write clearly what are the portion to check!!

set is open if. Rough.

for each $n \in A$

$$\exists \epsilon > 0, \text{ s.t. } B_\epsilon(n) \subseteq A$$

$$B_\epsilon(n) = \{y : |y - n| < \epsilon\}$$

$n \in \mathbb{R} : f(n) > a$.

consider

$$f(n) > a$$

since $f(n)$ is continuous at $n = n_0$.

$$\forall \epsilon > 0, \exists \delta \text{ s.t. } |f(n) - f(n_0)| < \epsilon$$

$$f(y) > a, \quad f(n) > a, \quad |y - n| < \delta$$

consider $\epsilon = f(n) - a$ $|f(y) - f(n)| < \epsilon$

$$f(c^-) < f(n) - \epsilon < f(y) < f(n) + \epsilon < f(n)$$

U is monotone but not continuous discontinuity is jump.

consider c pt of discontinuity.

$$\Rightarrow f(c^-) < f(c^+)$$

consider set $U = \{x \in \mathbb{R} : f(x) > a\}$

suppose set only contains

if set contains non

for set to be open

$n \in$

$$n = 0$$

$$\exists \epsilon \in B_\epsilon\{n\} = 1$$

(or 2) - consider $\epsilon < 1$.

$(0, 2)$ is open.

if i consider $n \in (0, 2)$ i can find $0 < c < n < k < 2$

take $\epsilon : \min \{n - c, n - 1\}$

$$B_\epsilon(n) : y : d(y, n) < \epsilon$$

$$B_r(n) : M$$

$$f(n) - \epsilon < f(y) < f(n) + \epsilon$$

$$|y - n| < r$$

$$B_r(n) : y \in \mathbb{R}$$

s.t.